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PROJECT FLOW-FAST

WAREHOUSE OPERATIONAL EXCELLENCE

A 30-60-90 Day Strategy to Eliminate Friction Across Inbound, Picking & Inventory

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FULFILLMENT AT SCALE: BOTTLENECKS TO BREAKTHROUGH

API Holdings Regional Fulfillment Center (12,000 Orders/Day | 18,000 SKUs | 85 Inbound Shipments)

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PROBLEM STATEMENT

At a scale of **12,000 orders** and **18,000 SKUs** daily, severe friction across inbound delays (**9.5-hour TAT**), floor picking (**55 lines/hour**), and low inventory accuracy (**94%**) causes high labor costs, stockouts, and avoidable losses from expiry, damage, and wrong dispatches.

INBOUND TAT
9.5 Hrs
Dock bottleneck

PICK SPEED
55 Lines/Hr
High walking waste

INVENTORY ACCURACY
94%
6% mismatch & gap

DAILY ERROR TRIPS
1,800 Lines
Stalled picks/day

Root-Cause Flow Diagram

Stage 1: Inbound
85 INBOUND TRUCKS



Unscheduled docks
Manual check-in

Stage 2: Latency
9.5-HR DOCK LAG



Daytime stockouts
Locked stock

Stage 3: Inventory
SCATTERED STORAGE



94% accuracy
Near-expiry leaks

Stage 4: Picking
FLOOR FRICTION



High walking waste
Empty bin waits

Outcome: Losses
CRITICAL CUSTOMER IMPACT



Wrong dispatches
Returns & scrap

Diagnosis & Priorities | **Solutions & Impact** | **Roadmap & KPIs**

OPERATIONAL EXCELLENCE: LEVERS & QUANTIFIED VALUE CREATION

Targeted Low-CapEx Process Innovations Across Inbound, Floor Picking & Inventory Governance

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UNIFIED ENGINE

1. INBOUND INTAKE

Dock Appointment Slots
Spread 85 loads smoothly across the day

ASN Pre-Receipt
Pre-printed pallet tags for direct scan-in

Green Channel QC
Fast-track trusted tier-1 pharma suppliers

2. VELOCITY PICKING

Fast-Mover ABC Zone
Top 20% SKUs placed near packing area

Multi-Order Batch Carts
Pick 8–12 customer orders per single loop

Auto-Trigger Refills
Auto-replenish forward bins at 30% buffer

3. CLOSED-LOOP FEFO

System-Locked FEFO Barcode
Block newer batches until older stock clears

Dynamic Cycle Counting
Trigger counts immediately on bin mismatch

Pack-Station Verification
100% final barcode check before dispatch

TAT: 9.5h ➔ 3.5h

Pick: 55 ➔ 95+ L/h

IMPACTS

1. DIRECT LABOR PRODUCTIVITY

- **Pick Rate:** 55 ➔ **95+ Lines/Hour**
- **Labor Freed:** 230 Hours/Day

💰 *Value: ₹8.27 Lakhs / Month*

2. REVENUE RECOVERY (INBOUND)

- **TAT:** 9.5 Hrs ➔ **≤ 3.5 Hours**
- **Stockouts Avoided:** **~180 Orders/Day**

💰 *Profit Margin: ₹4.86 Lakhs / Month*

3. DEFECT & EXPIRY REDUCTION

- **Expiry Scrap Cut:** **40% (FEFO)**
- **Mispicks Reduced:** **80%**

💰 *Savings: ₹15.12 Lakhs / Month*

4. WORKING CAPITAL RELEASE

- **Buffer Stock Cut:** **5%**
- **Cash Liquidity Freed:** **₹75 Lakhs**

💰 *Carrying Gain: ₹0.75 Lakhs / Month*

TOTAL VALUE DELIVERED: ~₹29.0 LAKHS / MONTH (₹3.48 CRORE ANNUALIZED RUN RATE) • ZERO HIGH CAPEX

30-60-90 Day Phased Rollout, Core KPIs & Operational Guardrails

Delivering Scalable Improvements with Clear Floor Governance and Defect Defenses

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PHASE 1: DAYS 1–30

DIAGNOSE & STABILIZE

- **ABC Velocity Cut:** Classify all 18,000 SKUs; isolate top 2,000 Class A medicines
- **Time-Motion Study:** Audit picker travel paths and dock check-in delays
- **Bin Baseline:** Clean up shelf labels and launch daily counts on fast movers

PHASE 2: DAYS 31–60

PILOT & RE-ENGINEER

- **Forward Zone Move:** Reslot top 2,000 SKUs directly next to pack stations
- **Dock Scheduling Pilot:** Roll out 1-hour booking slots for top 20 suppliers
- **WMS FEFO Lock:** Activate scanner hard stop preventing newer batch scans
- **Batch Cart Trial:** Test multi-order picking carts with a core 10-picker team

PHASE 3: DAYS 61–90

SCALE & SYSTEMATIZE

- **Warehouse-Wide Batching:** Expand multi-order batch carts across all zones
- **Auto-Refill Triggers:** System auto-alerts bulk replenishment at **30%** bin buffer
- **Vendor Scorecards:** Track ASN compliance, on-time arrivals, and near-expiry stock
- **SOP Institutionalization:** Standardize shift handover boards and count routines

Key Metric Scorecards

INBOUND DOCK-TO-STOCK TAT

Baseline: 9.5 Hours
90-Day Target: ≤ 3.5 Hours
🟢 *63% Faster Availability*



PICKER PRODUCTIVITY

Baseline: 55 Lines/Hour
90-Day Target: 95+ Lines/Hour
🟢 *+72% Throughput Surge*



INVENTORY RECORD ACCURACY

Baseline: 94.0%
90-Day Target: ≥ 99.2%
🟢 *Elimination of Bin Discrepancies*



DISPATCH ACCURACY & SCRAP

Baseline: High Expiry & Returns
90-Day Target: < 0.25% Scrap / ≥ 99.8% Accuracy
🟢 ***Zero-Mispick Dispatches***



🛡️ OPERATIONAL SAFEGUARDS:

- **Inbound:** **30%** Flex-Dock reserved for unscheduled trucks
- **FEFO Lock:** 2-tap supervisor override triggers an instant bin cycle-count
- **Demand Spikes:** Weekly automated Sunday re-scoring of Class A forward slots

Appendix

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Master Metric Transformation Summary

Metric Area	Baseline Metric PDF+ 1	Post-Solution Target PNG+ 1	Direction & Delta	Why It Changes (Operational Driver)
Inbound Dock-to-Stock TAT	9.5 Hours	**<= 3.5 Hours**	🔻 **63% Decrease**	Dock slot scheduling spreads 85 trucks evenly; ASN pre-barcoding enables instant scan-to-stock.
Picker Productivity	55 Lines/Hour	**95+ Lines/Hour**	🔺 **72% Increase**	ABC forward-pick zone cuts walking distance; batch carts pick 8–12 orders per loop.
Direct Labor Hours Needed	~545 Hours/Day	**~316 Hours/Day**	🔻 **42% Decrease**	Faster picking saves 230 labor hours daily, freeing up ~28 full-time warehouse associates.
Inventory Record Accuracy (IRA)	94.00%	**>= 99.2%**	🔺 **5.2% Increase**	Dynamic perpetual cycle counting and barcode-locked putaway eliminate stock discrepancies.
Daily Problematic Picks	1,800 lines/day	**< 240 lines/day**	🔻 **87% Decrease**	High inventory accuracy stops workers arriving at missing, mismatched, or empty bins.
Expiry & Scrap Loss	~1.0% of Inventory	**< 0.25% GMV**	🔻 **40%–75% Decrease**	System-enforced FEFO scanner lock prevents pickers from bypassing older stock.
Order Mispicks & Wrong Dispatches	~1.5% of Orders	**< 0.15% (>99.8%)**	🔻 **80%–90% Decrease**	Overhead pack-station barcode scanning verifies 100% of SKUs before shipping labels print.
Buffer Safety Stock	100% Baseline	**95% Baseline**	🔻 **5% Decrease**	Shorter inbound lead time and higher data accuracy reduce required safety buffer.
Cash Liquidity Freed	₹0 Baseline	**₹75 Lakhs Released**	🔺 **₹75L One-Time Unlock**	Lower safety inventory releases working capital directly into company cash reserves.

Appendix

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1. Direct Labor Productivity

Current Volume: 12,000 orders/day × 2.5 lines/order = 30,000 lines/day
Current productivity: 30,000 ÷ 55 = 545 hours/day
Future productivity: 30,000 ÷ 95 = 316 hours/day
Labor hours saved: 545 – 316 = 230 hours/day
Monthly savings: 230 × 30 × ₹120 = ₹8.28 lakh/month

2. Revenue & Availability Improvement

Lost orders due to stock unavailability: 12,000 × 3% = 360 orders/day
Recovered orders: 360 × 50% = 180 orders/day
Recovered GMV: 180 × ₹600 = ₹1.08 lakh/day
Monthly recovered GMV: ₹1.08 lakh × 30 = ₹32.4 lakh/month
Monthly contribution (15% margin): ₹32.4 lakh × 15% = ₹4.86 lakh/month
Dock-to-stock time: 9.5 hours → 3.5 hours = 6 hours faster

3. Scrap, Expiry & Return Reduction

A. Expiry Scrap Savings

Monthly inventory value: ₹21.6 crore
Current expiry losses: ₹21.6 crore × 1% = ₹21.6 lakh/month
Savings from 40% reduction: ₹21.6 lakh × 40% = ₹8.64 lakh/month

B. Mispick & Return Savings

Wrong orders: 12,000 × 1.5% = 180 orders/day
Errors prevented: 180 × 80% = 144 orders/day
Monthly savings: 144 × ₹150 × 30 = ₹6.48 lakh/month
Total Defect Savings: ₹8.64 lakh + ₹6.48 lakh = ₹15.12 lakh/month

4. Working Capital Optimization

Current inventory: ₹15 crore
Reduction in safety stock: ₹15 crore × 5% = ₹75 lakh released
Annual carrying cost: ₹75 lakh × 12% = ₹9 lakh/year
Monthly carrying cost savings: ₹9 lakh ÷ 12 = ₹75,000/month
Stock accuracy: 94% → 99.2%

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Thank You !!





